

Questions with Answer Keys

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Q1: 16 March (Shift 1) - Single Correct

The range of $a \in \mathbb{R}$ for which the function

$$f(x) = (4a - 3)(x + \log_e 5) + 2(a - 7) \cot\left(\frac{x}{2}\right) \sin^2\left(\frac{x}{2}\right)$$

$x + 2n\pi, n \in \mathbb{N}$, has critical points, is

(1) $(-3, 1)$

(2) $\left[-\frac{4}{3}, 2\right]$

(3) $[1, \infty)$

(4) $(-\infty, -1]$

Q2: 16 March (Shift 1) - Numerical

If the normal to the curve $y(x) = \int_0^x (2t^2 - 15t + 10) dt$ at a point (a, b) is parallel to the line $x + 3y = -5, a > 1$, then the value of $|a + 6b|$ is equal to _____

Q3: 16 March (Shift 2) - Single Correct

Let C be the locus of the mirror image of a point on the parabola $y^2 = 4x$ with respect to the line $y = x$. Then the equation of tangent to C at $P(2, 1)$ is :

(1) $x - y = 1$

(2) $2x + y = 5$

(3) $x + 3y = 5$

(4) $x + 2y = 4$

Q4: 16 March (Shift 2) - Single Correct

Let f be a real valued function, defined on

$\mathbb{R} - \{-1, 1\}$ and given by

$$f(x) = 3 \log_e \left| \frac{x-1}{x+1} \right| - \frac{2}{x-1}$$

Then in which of the following intervals, function $f(x)$ is increasing?

Questions with Answer Keys

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(1) $(-\infty, -1) \cup \left(\left[\frac{1}{2}, \infty\right) - \{1\}\right)$

(2) $(-\infty, \infty) - \{-1, 1\}$

(3) $\left(-1, \frac{1}{2}\right]$

(4) $\left(-\infty, \frac{1}{2}\right] - \{-1\}$

Q5: 17 March (Shift 1) - Single Correct

Which of the following statements is correct for the function $g(\alpha)$ for $\alpha \in \mathbb{R}$ such that

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x} dx$$

(1) $g(\alpha)$ is a strictly increasing function(2) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$ (3) $g(\alpha)$ is a strictly decreasing function(4) $g(\alpha)$ is an even function

Q6: 17 March (Shift 2) - Single Correct

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} \left(2 - \sin\left(\frac{1}{x}\right)\right)|x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then f is(1) monotonic on $(-\infty, 0) \cup (0, \infty)$ (2) not monotonic on $(-\infty, 0)$ and $(0, \infty)$ (3) monotonic on $(0, \infty)$ only(4) monotonic on $(-\infty, 0)$ only

Q7: 17 March (Shift 2) - Numerical

Let $f: [-1, 1] \rightarrow \mathbb{R}$ be defined as $f(x) = ax^2 + bx + c$ for all $x \in [-1, 1]$, where $a, b, c \in \mathbb{R}$ such that $f(-1) = 2$, $f'(-1) = 1$ and for $x \in (-1, 1)$ the maximum value of $f''(x)$ is $\frac{1}{2}$. If $f(x) \leq \alpha$ $x \in [-1, 1]$, then the least value of α is equal to _____.

Q8: 18 March (Shift 2) - Single Correct

Let a tangent be drawn to the ellipse $\frac{x^2}{27} + y^2 = 1$ at $(3\sqrt{3} \cos \theta, \sin \theta)$ where $\theta \in \left(0, \frac{\pi}{2}\right)$. Then the value of θ such that the sum of intercepts on axes made by this tangent is minimum is equal to:

(1) $\frac{\pi}{8}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{6}$

(4) $\frac{\pi}{3}$

Q9: 18 March (Shift 2) - Numerical

Let $P(x)$ be a real polynomial of degree 3 which vanishes at $x = -3$. Let $P(x)$ have local minima at $x = 1$, local maxima at $x = -1$ and $\int_{-1}^1 P(x) dx = 18$, then the sum of all the coefficients of the polynomial $P(x)$ is equal to ____

Answer Key**Q1 (2)****Q2 (406)****Q3 (1)****Q4 (1)****Q5 (4)****Q6 (2)****Q7 (5)****Q8 (3)****Q9 (8)**

Q1 (2)

$$f(x) = (4a - 3)(x + \log_e 5) + (a - 7) \sin x$$

$$f(x) = (4a - 3)(1) + (a - 7) \cos x = 0$$

$$\Rightarrow \cos x = \frac{3-4a}{a-7}$$

$$\frac{3a+4}{a-7} \leq 0$$

$$\frac{3-4a}{a-7} + 1 \geq 0$$

$$\frac{3-4a+a-7}{a-7} \geq 0$$

$$\frac{-3a-4}{a-7} \geq 0$$

$$\frac{3-4a}{a-7} < 1$$

$$\frac{3-4a}{a-7} - 1 < 0$$

$$\frac{3-4a-a+7}{a-7} < 0$$

$$\frac{5a+10}{a-7} < 0$$

$$\frac{5a-10}{a-7} > 0$$

$$\frac{5(a-2)}{a-7} > 0$$



$$\alpha \in \left[-\frac{4}{3}, 2\right)$$

$$\text{Check end point } \left[-\frac{4}{3}, 2\right)$$

Q2 (406)

$$y(x) = \int_0^x (2t^2 - 15t + 10) dt$$

$$y'(x)|_{x=a} = [2x^2 - 15x + 10]_a = 2a^2 - 15a + 10$$

$$\text{Slope of normal} = -\frac{1}{3}$$

$$\Rightarrow 2a^2 - 15a + 10 = 3 \Rightarrow a = 7$$

$$\& a = \frac{1}{2} \text{ (rejected)}$$

$$b = y(7) = \int_0^7 (2t^2 - 15t + 10) dt$$

$$= \left[\frac{2t^3}{3} - \frac{15t^2}{2} + 10t \right]_0^7$$

$$\Rightarrow 6b = 4 \times 7^3 - 45 \times 49 + 60 \times 7$$

$$|a + 6b| = 406$$

Q3 (1)

Hints and Solutions

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Given $y^2 = 4x$

Mirror image on $y = x \Rightarrow C : x^2 = 4y$

$$2x = 4 \cdot \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2}$$

$$\left. \frac{dy}{dx} \right|_{P(2,1)} = \frac{2}{2} = 1$$

Equation of tangent at (2,1)

$$\Rightarrow y - 1 = 1(x - 2)$$

$$\Rightarrow x - y = 1$$

Q4 (1)

$$f(x) = 3\ln(x-1) - 3\ln(x+1) - \frac{2}{x-1}$$

$$f'(x) = \frac{3}{x-1} - \frac{3}{x+1} + \frac{2}{(x-1)^2}$$

$$f'(x) = \frac{4(2x-1)}{(x-1)^2(x+1)}$$

$$f'(x) \geq 0$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[\frac{1}{2}, 1\right) \cup (1, \infty)$$

Q5 (4)

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\pi/3} \frac{\sin^\alpha x}{(\sin^\alpha x + \cos^\alpha x)} \dots (i)$$

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\pi/3} \frac{\cos^\alpha x}{(\sin^\alpha x + \cos^\alpha x)} \dots (ii)$$

$$(1) + (2)$$

$$2g(\alpha) = \frac{\pi}{6}$$

$$g(\alpha) = \frac{\pi}{12}$$

Constant and even function

Due to typing mistake it must be bonus.

Q6 (2)

$$f(x) = \begin{cases} -x \left(2 - \sin\left(\frac{1}{x}\right) \right) & x < 0 \\ 0 & x = 0 \\ x \left(2 - \sin\left(\frac{1}{x}\right) \right) & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} - \left(2 - \sin\frac{1}{x} \right) - x \left(-\cos\frac{1}{x} \cdot \left(-\frac{1}{x^2} \right) \right) & x < 0 \\ \left(2 - \sin\frac{1}{x} \right) + x \left(-\cos\frac{1}{x} \cdot \left(-\frac{1}{x^2} \right) \right) & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} -2 + \sin\frac{1}{x} - \frac{1}{x}\cos\frac{1}{x} & x < 0 \\ 2 - \sin\frac{1}{x} + \frac{1}{x}\cos\frac{1}{x} & x > 0 \end{cases}$$

$f'(x)$ is an oscillating function which is non-monotonic in $(-\infty, 0) \cup (0, \infty)$

Q7 (5)

$$f : [-1, 1] \rightarrow \mathbb{R}$$

$$f(x) = ax^2 + bx + c$$

$$f(-1) = a + b + c = 2$$

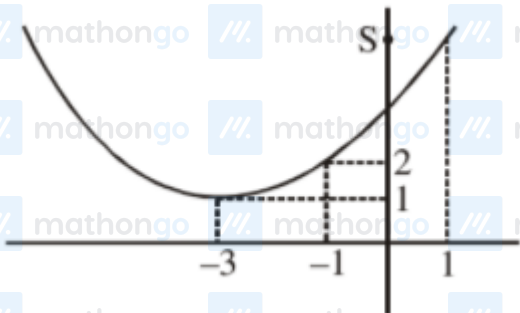
$$f'(-1) = -2a + b = 1$$

$$f''(x) = 2a$$

$$\Rightarrow \text{Max. value of } f''(x) = 2a = \frac{1}{2}$$

$$\Rightarrow a = \frac{1}{4}; \quad b = \frac{3}{2}; \quad c = \frac{13}{4}$$

$$\therefore f(x) = \frac{x^2}{4} + \frac{3}{2}x + \frac{13}{4}$$



Hints and Solutions

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For, $x \in [-1, 1] \Rightarrow 2 \leq f(x) \leq 5$

$\therefore$ Least value of α is 5

Q8 (3)

Equation of tangent be

$$\frac{x \cos \theta}{3\sqrt{3}} + \frac{y \sin \theta}{1} = 1, \quad \theta \in \left(0, \frac{\pi}{2}\right)$$

intercept on x-axis

$$OA = 3\sqrt{3} \sec \theta$$

intercept on y-axis

$$OB = \operatorname{cosec} \theta$$

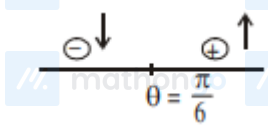
Now, sum of intercept

$$= 3\sqrt{3} \sec \theta + \operatorname{cosec} \theta = f(\theta) \text{ let}$$

$$f'(\theta) = 3\sqrt{3} \sec \theta \tan \theta - \operatorname{cosec} \theta \cot \theta$$

$$= 3\sqrt{3} \frac{\sin \theta}{\cos^2 \theta} - \frac{\cos \theta}{\sin^2 \theta}$$

$$= \underbrace{\frac{\cos \theta}{\sin^2 \theta}}_{\oplus} \cdot 3\sqrt{3} \left[\tan^2 \theta - \frac{1}{3\sqrt{3}} \right] = 0 \Rightarrow \theta = \frac{\pi}{6}$$



$\Rightarrow$ at $\theta = \frac{\pi}{6}$, $f(\theta)$ is minimum

Q9 (8)

$$\text{Let } p'(x) = a(x-1)(x+1) = a(x^2-1)$$

$$p(x) = a \int (x^2-1) dx + c$$

$$= a \left(\frac{x^3}{3} - x \right) + c$$

$$\text{Now } p(-3) = 0$$

$$\Rightarrow a \left(-\frac{27}{3} + 3 \right) + c = 0$$

$$\Rightarrow -6a + c = 0$$

Hints and Solutions

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$$\text{Now } \int_{-1}^1 \left(a \left(\frac{x^3}{3} - x \right) + c \right) dx = 18$$

$$= 2c = 18 \Rightarrow c = 9 \quad \dots (2)$$

 $\Rightarrow$ from

$$(1) \& (2) \Rightarrow -6a + 9 = 0 \Rightarrow a = \frac{3}{2}$$

$$\Rightarrow p(x) = \frac{3}{2} \left(\frac{x^3}{3} - x \right) + 9$$

sum of coefficient

$$= \frac{1}{2} - \frac{3}{2} + 9$$

$$= 8$$